# Teacher Wu Kuan | See It at a Glance #5 — Pythagorean Theorem by Area

English lecture notes · YouTube Short f47RMUhjKs8

## 1. The trap question

Why $a^2 + b^2 = c^2$?

## 2. The method

- Four copies of the triangle in a $(a + b) \times (a + b)$ square, arranged two ways.
- Leftover area: $c^2$ in one arrangement, $a^2 + b^2$ in the other.
- So $a^2 + b^2 = c^2$ ($c$ is the side opposite the right angle).

## 3. Parent check method

Ask your child to say which side is c before plugging in numbers.

## Practice

1. Legs 6 and 8. Find the hypotenuse.
2. Hypotenuse 13, one leg 5. Find the other leg.
3. Legs 1 and 1. Find the hypotenuse.
4. Is a triangle with sides 7, 24, 25 a right triangle?
5. A screen is 30 in by 40 in. Find its diagonal.

## Answers

1. 10
2. 12
3. $\sqrt{2}$
4. Yes ($49 + 576 = 625$)
5. 50 in

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